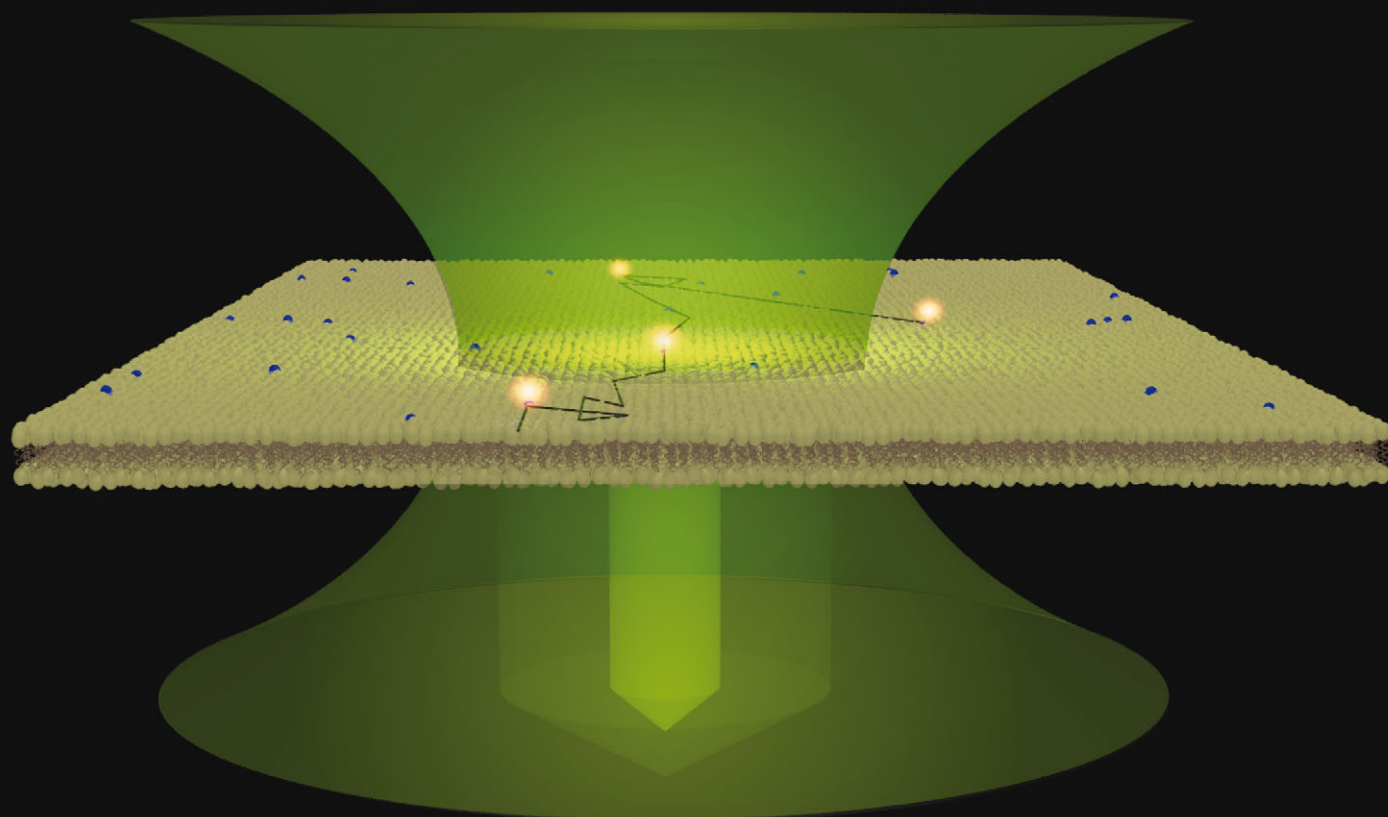


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COVER ARTICLE

J. Ortega-Arroyo and P. Kukura
Interferometric scattering
microscopy (iSCAT): new frontiers in
ultrafast and ultrasensitive optical
microscopy

HOT ARTICLE

Zeuch *et al*
Sulfur dioxide oxidation induced
mechanistic branching and particle
formation during the ozonolysis of
beta-pinene and 2-butene



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PERSPECTIVE

Interferometric scattering microscopy (iSCAT): new frontiers in ultrafast and ultrasensitive optical microscopy

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Optical microscopes have for centuries been our window to the microscopic world. The advent of single-molecule optics over the past few decades has ushered in a new era in optical imaging, partly because it has enabled the observation of motion and more recently structure on the nanoscopic scale through the development of super-resolution techniques. The large majority of these studies have relied on the efficient detection of fluorescence as the basis of single-molecule sensitivity. Despite the many advantages of using single emitters as light sources, the intensity and duration of their emission impose fundamental limits on the imaging speed and precision for tracking studies. Here, we discuss the potential of a novel imaging technique based on interferometric scattering (iSCAT) that pushes both the sensitivity and time resolution far beyond what is currently achievable by single-emitter-based approaches. We present recent results that demonstrate single-molecule sensitivity and imaging speeds on the microsecond timescale.

1. Introduction

Single-particle tracking (SPT) is a tremendously useful technique to study a variety of processes in the life sciences and condensed matter physics. Time scales range over several orders of magnitude from nanoseconds, as in the recent observation of ballistic motion,¹ to seconds in nanometric single-molecule localisation used to decipher the stepping mechanism of molecular motors.^{2–6}

Broadly speaking, SPT has two major roles in contemporary science: on the one hand it serves as a first step towards making movies on the nanoscale by allowing us to observe dynamic motion of individual particles or even molecules. On the other hand, it allows us to use the motion of small objects as a probe for nanoscopic potentials. In this respect, the non-invasiveness of optical imaging is a distinct advantage over surface-sensitive techniques such as atomic force microscopy, although at the expense of limited optical resolution.

The basic idea behind SPT is illustrated using the diffusion of lipids in a membrane in Fig. 1a. Here, a labelled receptor

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Philipp Kukura

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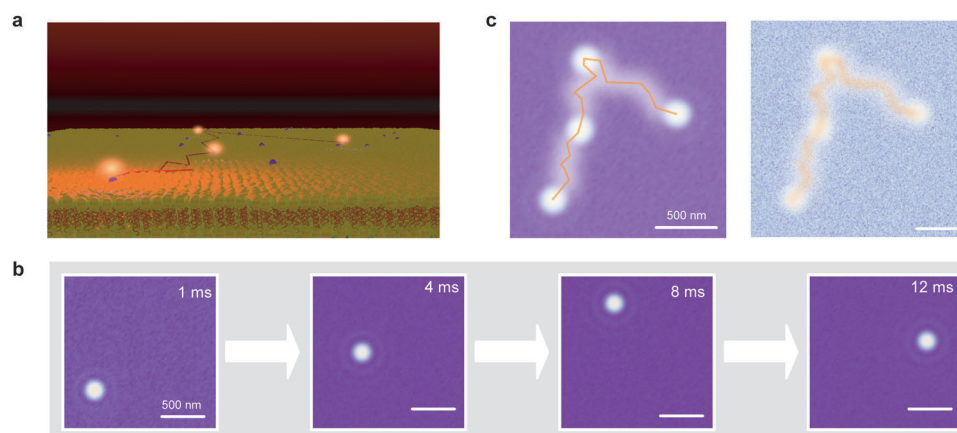


Fig. 1 (a) Illustration of a typical single-particle tracking assay on a membrane bilayer with receptors depicted in purple and fluorescent markers in bright orange. The trajectory of a labelled receptor as it diffuses across the membrane, represented as the solid red line, can be followed by tracing the fluorescence signal produced by the marker. (b) For labels smaller than the diffraction limit, the signal obtained at the detector is that of a PSF resembling a 2D Gaussian, which moves as a function of time. All scale bars correspond to 500 nm. (c) Reconstructed trajectory of the motion of the particle PSF in (b). Multiple images are overlaid to show the motion of the PSF over time. The true trajectory of the receptor (orange line) can be reconstructed with a greater precision than the size of the PSF by finding the centre of mass of the particle at each time point. To the far right, a lower SNR version of the same image sequence translates into a deterioration of the precision with which the trajectory can be reconstructed.

carries a fluorescent marker as it skates through the liquid membrane. For a nanoscopic label, its optical image, known as the point spread function (PSF), closely resembles that of a 2D Gaussian function with a full-width at half-maximum of roughly half the wavelength of light as dictated by Abbe's diffraction limit. As the receptor diffuses, the position of the PSF moves as a function of time (Fig. 1b). Despite the label's small size, its PSF, and thus its centre of mass is very well defined and can be determined using several approaches^{7,8} such as a fit to a 2D Gaussian function.⁹ Repeating the procedure for consecutive images reveals a detailed nanoscopic trajectory of the receptor motion with a precision far exceeding the diffraction limit, thereby resulting in a much more detailed picture of the object's motion (Fig. 1c). From a pure tracking perspective, the precision and speed with which the motion of the label can be followed depend solely on the signal-to-noise ratio (SNR).^{10–12} Thus, as the image quality deteriorates due either to fluctuations caused by shot noise or the appearance of background noise, the centre of mass becomes less well defined and the corresponding trajectory increasingly noisier, leading to the eventual loss of most of the detail of the motion.

The major difficulty when dealing with single emitters for tracking studies arises from two limitations. Firstly, they exhibit unique photophysical and chemical effects, such as photoblinking and bleaching, which limit the total achievable photon yield.^{13,14} Solutions for these effects are materialising in the form of non-blinking quantum dots¹⁵ and NV-centres in nanodiamonds,^{16,17} although routine incorporation of these labels in biological environments remains a challenge. More importantly and fundamentally, however, all of these systems are single quantum emitters and thus only capable of emitting a certain maximum number of photons per unit time.¹⁸ This arises from the fact that emission of a photon involves prior population of an excited electronic state, which has a lifetime of a few to tens of nanoseconds. In a realistic scenario, all of

the above effects limit achievable excitation rates to roughly 1 MHz which, in combination with the typical losses in current state-of-the-art optical microscopes, result in a rough rule of achievable single emitter localisation and speed:

$$\sigma(\text{time, space}) = 1 \text{ nm Hz}^{-1/2}$$

The implications of this equation are quite straightforward and have been verified in all state of the art single emitter tracking experiments over the past decade.^{2,18–23} For imaging at 1 Hz, localisation with 1 nm precision is possible, while at 100 Hz the precision drops off to 10 nm. At 100 kHz only about one photon will be detected per image, corresponding to a SNR of 1 and thus no improvement in localisation over the diffraction limit. At this level, all information on nanoscopic motion is lost.

These limitations would not pose a problem if motion on the nanoscale took place on the timescale of seconds or slower, but unfortunately, nature is several orders of magnitude faster. As an example, the diffusion coefficient for motion of lipids in bilayer membranes is on the order of $1 \mu\text{m}^2 \text{s}^{-1}$, which translates into a root-mean-square displacement of 2 nm in just one microsecond.²⁴ The corresponding root-mean-square displacements for a 10 nm protein in a cellular environment (9 nm) and water (16 nm) are even less encouraging if the goal is to follow the fast dynamics of the protein and its environment. Taken together, single-emitter tracking is immensely powerful because it allows us to follow the motion of individual species, but fails fundamentally in terms of revealing nanoscopic dynamics in real time.

Due to this spectacular discrepancy between what one would like to achieve and what one can actually achieve with single emitters, there has been extensive work towards identifying and applying alternative contrast mechanisms that would allow not only precise, but simultaneously fast nanoscopic tracking. In this perspective, we will focus on light

scattering and in particular interferometric scattering microscopy (iSCAT), a novel technique that is capable of imaging very small labels at very high speeds. We provide a detailed comparison with both pure scattering and fluorescence microscopy and demonstrate the unique capabilities of iSCAT with respect to sensitivity and speed using recent experimental results.

2. Scattering as an alternative to fluorescence detection

In contrast to fluorescence where the achievable photon flux is fundamentally capped due to optical saturation, in light scattering it is only limited by the incident photon flux. Since a nano-object will always scatter a given percentage of the incident light as determined by the ratio of the scattering cross section over the incident beam size, one can adjust the scattered photon flux simply by tuning the incident power. For a perfect experiment, there are thus no boundaries for the achievable combination of temporal and spatial precision because one can always obtain the required scattered photon flux by increasing the incident power. Drawing the analogy between a light scattering nano-object and a “nano-mirror” implies that as long as the object is illuminated, indefinite observation times are possible without being subjected to phenomena such as photobleaching and blinking.

Based on these arguments, one might infer that light scattering is indeed far superior to fluorescence detection due to its never-ending source of photons. In practice, however, background scattering poses a significant challenge towards the limits of detection and localization of nano-objects. As an example, consider the detection of light scattered by a relatively large, 160 nm silica bead. The corresponding scattering cross section in water amounts to $\sim 3 \times 10^{-12} \text{ cm}^2$ which in comparison to the minimal diffraction limited beam size at 532 nm of $3 \times 10^{-10} \text{ cm}^2$ results in about 1% of the incident photons being scattered by the particle. For nm localisation precision, SNRs of 30–60 are required,^{10,11,20} meaning that any spurious reflections have to be eliminated up to a level of 0.02% of the incident light level once the limited collection efficiency of microscope objectives is taken into account. The difficulty associated with this task is evidenced by considering that a standard high numerical aperture microscope objective consists of several lenses that will reflect at least 0.1% of the incident light per surface even in the presence of excellent anti-reflection coatings. Given that there are many additional sources of spurious reflections such as dust particles, inclusions in microscope immersion oil, surface inhomogeneities and refractive index changes such as at the glass–water interface, the challenge of detecting pure light scattering with high SNRs becomes evident. The situation becomes even worse and effectively non-manageable if measurements are to be performed in intrinsically scattering environments such as the inside of cells.

2.1 Pure scattering detection

Optimising the SNR necessarily involves maximising the signal and minimising any background noise. The latter is best achieved using dark-field microscopy where the incident light is excluded from the light scattered by the sample. The most successful approaches take advantage of total internal

reflection (TIR) at a refractive index gradient either using objective-TIR or prism-TIR as shown in Fig. 2a. Such a TIR-setup uses a prism such that the incident laser light is directed to the sample at a steep enough angle to be fully reflected at the water–coverglass interface. As a result, an exponentially decaying evanescent field (E_e) is produced within the sample region with a depth of a few hundred nanometres.^{25,26} Forward scattered light (E_s) from the particles within this evanescent field region is collected by a microscope objective and imaged onto a CCD camera. To further improve the collection of scattered light the experiment can be setup to collect back-scattered light.²⁷ In both cases, particles detected under dark-field appear as bright spots on top of a dark background (Fig. 2b).

The scattering efficiency depends critically on the interaction strength of the incident light with the nano-object. For particles that are small compared to the wavelength of light, the scattered field at the detector is given by $E_s = sE_i$ where $s = |s|e^{i\phi}$ scales with the polarisability α of the particle and ϕ is the scattering phase. Using general Mie theory, the scattering amplitude can be written as:

$$s(\lambda) = \eta\alpha(\lambda) = \eta\epsilon_{\text{med}}(\lambda)\pi\frac{D^3}{2}\frac{\epsilon_{\text{part}}(\lambda) - \epsilon_{\text{med}}(\lambda)}{\epsilon_{\text{part}}(\lambda) + 2\epsilon_{\text{med}}(\lambda)} \quad (1)$$

where $\epsilon_{\text{med/part}}$ are the complex dielectric constants of the medium and the particle, D is the particle diameter and η a

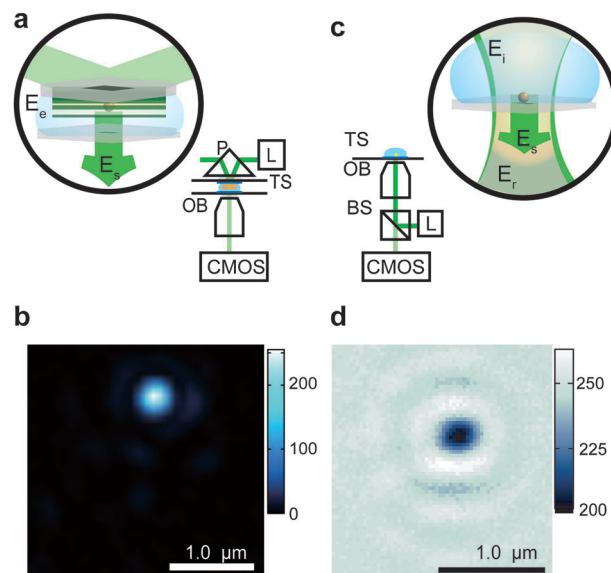


Fig. 2 (a) Representative schematic diagram for total internal reflection dark field microscopy (TIRDFM): prism (P), translation stage (TS), objective (OB), CMOS camera (CMOS). Inset, signal generation: objective collects the light forward scattered (E_s) by particles within the evanescent field (E_e) produced by total internal reflection. (b) Image of a 40 nm gold particle obtained by TIRDFM.²⁷ The particle appears as a bright spot over a dark background. Intensity scale in arbitrary units. (c) Schematic diagram for interferometric scattering microscopy: beamsplitter (BS). Inset, signal generation: objective tightly focuses the incident light (E_i) onto the sample and collects the light reflected (E_r) at the glass coverslip interface and the light backscattered by particles within the illumination region (E_s). (d) Analogous image of a 40 nm gold particle detected by iSCAT. The particle now appears as a dark spot over a constant bright background. Intensity scale in arbitrary units.

constant that takes into account the detection efficiency of the experimental setup. In general, the measured intensity at the detector is given by,

$$I_{\text{det}} = |E_b + E_s|^2 = |E_i|^2 \{b^2 + |s|^2 - 2b|s|\sin\phi\}, \quad (2)$$

where E_b denotes any background field and $|E_i|^2 b^2$ is the background intensity. In a dark-field experiment, b is minimised such that the dominant term reaching the detector is the pure scattering contribution, $|E_i|^2 |s|^2$. To maximise this signal, gold nanoparticles (AuNPs) are particularly useful, because they exhibit a strong plasmon resonance in water centred around 530 nm for spherical particles ranging in diameter from 10 to 100 nm. In terms of eqn (1) above, this resonance occurs when the expression $|\epsilon_{\text{part}}(\lambda) + 2\epsilon_{\text{med}}(\lambda)|$ is minimised, which results in an enhanced polarisability and thus larger scattering amplitudes and cross-sections. In the case of a gold nanoparticle immersed in water, $|\epsilon_{\text{part}}(\lambda) + 2\epsilon_{\text{med}}(\lambda)|$ reaches a minimum value of $-0.60 + 2.57i$, which corresponds to a magnitude of 2.64. In fact, a 40 nm AuNP illuminated at 532 nm exhibits roughly the same scattering cross section in water as the 160 nm silica bead, thereby enabling the use of a much smaller label for the same scattered photon flux. Despite the plasmonic enhancement, the pure scattering term scales as D^6 (see eqn (1)), meaning that a two-fold reduction in particle size leads to as much as a 64-fold decrease in the purely scattered signal. Even if most background light intensity is suppressed, the intrinsic limitation to any scattering detection scheme will be the presence of background scattering due to sample imperfections. In a realistic experiment, this means that dark-field studies are performed with AuNP diameters ≥ 40 nm. Below this threshold even the most refined dark-field approaches can neither suppress the background light sufficiently nor reduce the presence of sample imperfections to a significant extent to achieve a high SNR for nm localisation precision measurements.

2.2 Interferometric scattering detection

Although the effect of the label size on single-particle tracking experiments is not precisely known, it is highly desirable to use smaller sized labels, especially in the life sciences, where most of the biological nanomachinery ranges in size from a few to several tens of nanometres. The simple equations outlined above immediately suggest one approach that may be successful: detection of the third, interference term, in eqn (2). This term scales linearly with the scattered field amplitude and thus linearly with the polarisability and particle volume. Any such technique clearly has the massive advantage that reducing the particle size by a factor of 2 carries with it an almost order of magnitude smaller drop in the detected signal (from 64 to 8).

Today, there are several approaches that take advantage of this effect more or less directly. They include absorption-based methods,^{28,29} photothermal detection^{30,31} and interferometric techniques using external^{32,33} or common path reference waves.³⁴ Photothermal detection has the unique advantage of enabling the identification of resonant objects even in strongly scattering environments, while externally referenced interferometric detection allows fine-tuning of the phase difference, ϕ , and thus the signal magnitude. Common path interferometric scattering detection dates back to 1965 with

the technique known as interference reflection microscopy (IRM) developed by Curtis³⁵ and further optimized by Ploem with the introduction of the antiflex illumination scheme.³⁶ As a precursor to iSCAT, this detection technique was used to study cell adhesion^{37,38}, thicknesses of thin films over glass substrates,³⁹ interactions between large colloidal particles and surfaces,⁴⁰ reconstruction of interfacial topology,^{41,42} and phase separation in lipid bilayers.³⁹

In its most recent incarnation, common path interferometric scattering detection has the major advantage of being the only approach that operates in wide-field mode much in the same way as in fluorescence microscopy. It enables large fields of view (hundreds of μm^2) and recording speeds approaching the μs frontier with nm localisation precision. At the same time, the associated experimental setup is comparatively trivial. In fact, most current confocal microscopes are already performing iSCAT experiments everyday; they simply do not have a detector at the correct position to record the relevant signal.

To understand the basic operating principle of iSCAT, it is instructive to compare the experimental setup with that of dark-field microscopy. Rather than trying to eliminate background light, in iSCAT it is explicitly collected and sent to the detector together with any scattered light. Thus, a typical setup resembles an inverted microscope where incident light (E_i) from a laser source is tightly focused onto the sample by a high numerical aperture objective (Fig. 2c). A portion of the scattered light together with the incident light partially reflected at the glass/water interface are collected by the objective and returned towards a 50 : 50 beamsplitter. Half of this light passes through the beamsplitter and reaches the detector resulting in a detected intensity very much like the one described above, except that the background light field (E_b) has been replaced by the reflected light field (E_r):

$$I_{\text{det}} = |E_r + E_s|^2 = |E_i|^2 \{r^2 + |s|^2 - 2r|s|\sin\phi\}, \quad (3)$$

For the glass–water interface $r \sim 0.065$, which is much greater than $|s|$ for particles with $D < 50$ nm. The signal contrast can then be written as:

$$\frac{I_{\text{particle}}}{I_{\text{background}}} = 1 - \frac{2r|s|\sin\phi}{r^2}. \quad (4)$$

There are two critical observations associated with this equation: firstly, the signal contrast now scales linearly with the scattering amplitude and thus particle volume. This reduction in size-dependence makes it possible to detect AuNPs as small as 5 nm,⁴³ almost an order of magnitude smaller than what is possible with dark-field techniques. Secondly, particles appear dark on top of a large but constant background rather than as bright spots (Fig. 2d).

These considerations nicely illustrate why interferometric detection makes it easier to image small particles in principle. Nevertheless, all of the arguments made above also apply to any background scatterers such as surface variations on the nanoscale. Roughness can be thought of as tiny, randomly shaped nanoparticles made of glass sitting on top of an atomically flat glass surface. Since their refractive index differs from that of the surrounding water, they generate a scattering signal according to eqn (2). In fact, exactly this type of

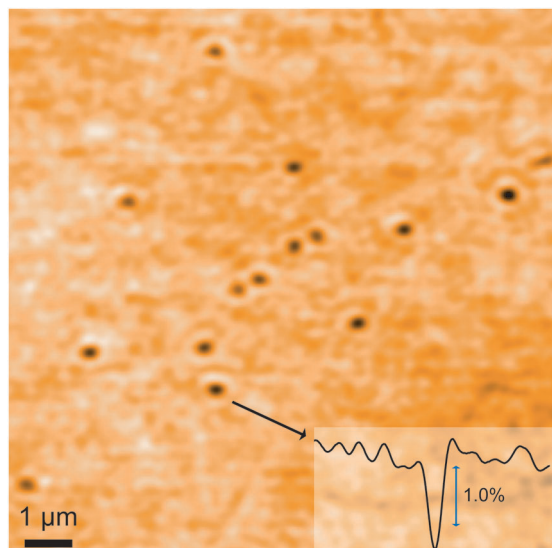


Fig. 3 Image of 15 nm Au particles spin coated onto the surface of a clean glass coverslip immersed in water obtained by illumination at $\lambda = 532$ nm with iSCAT microscopy. Variations in the background correspond to scattering signal contributions from tiny imperfections in the glass surface. A cross section of the image is shown to indicate the extinction in the signal produced by a single Au particle compared to the glass surface roughness.

behaviour is observed in Fig. 3: while the 15 nm AuNPs are clearly visible, there is also a noticeable background roughness that is reproducible and therefore not noise. In this respect, interferometric detection is as equally non-background free as dark-field scattering detection.

The important distinction between dark-field and iSCAT arises from the fact that in interferometric detection the strong reflection at the sample surface overwhelms any other spurious reflections. In addition, because these contributions are constant, they can be easily subtracted with a median filter⁴⁴ leaving only features caused by the sample itself. In terms of the equations above we could rewrite them as:

$$\begin{aligned} I_{\text{dark-field}} &= I_{\text{signal}} + I_{\text{b}} \\ &= |E_i|^2 \{r^2 + |s|^2 - 2r|s|\sin \phi\} + I_{\text{b}} \\ &= |E_i|^2 |s|^2 + I_{\text{b}} \end{aligned} \quad (5)$$

assuming $r \ll s$ for a strong scatterer and efficient incident light rejection. For interferometric detection, however:

$$\begin{aligned} I_{\text{iSCAT}} &= I_{\text{signal}} + I_{\text{b}} \\ &= |E_i|^2 \{r^2 + |s|\sin \phi\} + I_{\text{b}} \\ &= |E_i|^2 \{r^2 - 2r|s|\sin \phi\} \end{aligned} \quad (6)$$

for $I_{\text{reflected}} \gg I_{\text{background}}$ and $r \gg s$. For strong scatterers in interferometric detection, the pure and interference terms compete, which leads to a switch from negative interferometric to positive pure scattering signal contrast as the size of the particle increases. This switch happens at different particle sizes depending on the reflectivity of the interface ranging from 80 nm in air to about 40 nm in water as r is decreased.³⁴ In general, spurious reflections are negligible compared to the actual signal ($|E_i|^2 2r|s|\sin(\phi) > I_{\text{background}}$) as well as the large

constant background, r^2 , and thus no longer deteriorate the image quality even when $|s|$ becomes small. In this way, the only remaining background is due to sample imperfections such as roughness. Therefore, in practice much smaller particles can be detected by interferometric scattering due to the dynamic range scaling of D^3 as opposed to D^6 for purely scattering detection. Importantly, the constant scattering background due to sample imperfections can be measured with very high precision because it can be averaged over extensive periods of time prior to the experiment. Thus, even after background subtraction there is negligible image degradation.

Despite the many advantages of interferometric scattering detection outlined above, several difficulties remain. Scanning the sample over the laser focus necessarily translates laser intensity fluctuations into background noise. This noise is difficult to suppress beyond 0.05% with commercially available laser systems, which suffer from intensity stabilization problems, thereby limiting the sensitivity. Similarly, total internal reflection observed with high numerical aperture microscope objectives restricts the usable numerical aperture below 1.3 and 1.0 for water and air, respectively. Finally and most importantly, sample scanning is necessarily slow, leading to image acquisition times on the order of a few to tens of seconds making the observation of any fast dynamics impossible.

The key to minimizing the effect of all of the above limitations is to switch from sample (Fig. 4a) to beam scanning (Fig. 4b) and from point to wide-field detection. In terms of the scattering contrast, the most important parameter is the ratio of scattering amplitude to the size of the diffraction-limited beam. Analogous to fluorescence, the excitation spot size in interferometric scattering is irrelevant for the final resolution obtained on a camera. After all, the pixels of the CCD camera do not provide any knowledge of how the sample was illuminated, but rather record a diffraction-limited image of the sample region. In this fashion, the objective can be under-filled to avoid total internal reflection without any loss in signal contrast. The first indication that such an approach is equally valid was reported by Jacobsen *et al.* using standard wide-field illumination by focusing the incident laser light into the back focal plane of the

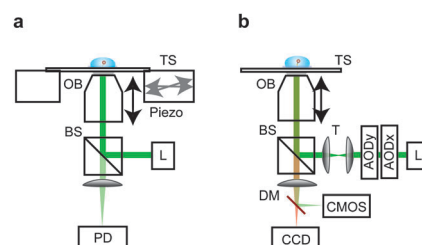


Fig. 4 (a) Experimental setup for interferometric scattering microscopy in confocal detection mode using a point-like detector and a piezo translation stage to raster scan the sample. (b) State of the art experimental iSCAT setup in wide-field detection mode using CMOS cameras as detectors and acousto-optic deflectors (AODs) to scan the beam over the sample. Fluorescence signals (red beam path) can be detected by addition of a fluorescence channel with the use of a dichroic mirror in the imaging beam path and a CCD camera. Photodiode (PD), objective (OB), translation stage (TS), beam-splitter (BS), laser (L), acousto-optic deflector (AOD), telecentric lens system (T), dichroic mirror (DM).

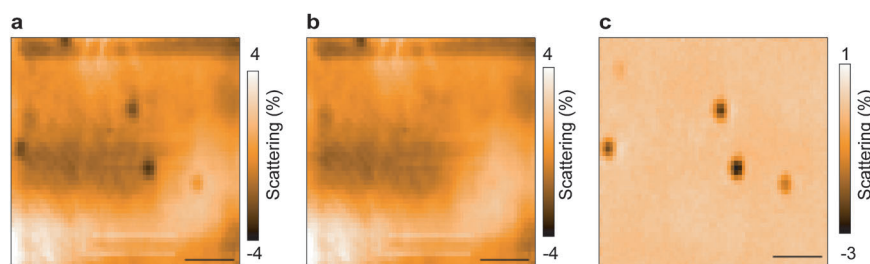


Fig. 5 (a) Representative raw interferometric scattering image of a recorded movie of SV40 particles bound to receptor GM1 diffusing on a supported lipid bilayer. (b) Background image obtained after median analysis of the raw movie sequence shown in (a). (c) Subtraction of (b) from (a). Scale bars, 1 μm .

microscope objective.⁴³ The quality of the resulting images, however, was rather poor due to strong reflections from within the microscope objective caused by the focused beam.

To avoid these difficulties, beam scanning can be performed at up to 100 kHz using acousto-optic deflectors with a collimated input beam.⁴⁵ By adjusting the *x*- and *y*-scanning frequencies to several tens of kHz, the sample is scanned over hundreds of times during the exposure time of a typical CCD camera running at full speed (a few to tens of ms). This creates a uniformly illuminated image without any interference fringes. Such an approach has the added advantage that the laser beam illuminates the same sample spot periodically at a fixed frequency resulting in a lock-in-like reduction of laser noise by narrowing down the detection bandwidth.

With this approach, it is straightforward to perform shot-noise-limited experiments at maximum numerical aperture and speeds limited only by the frame rate of the CCD camera. Eqn (6) above demonstrates that the iSCAT signal is detected as a dip on top of a large background. The SNR thus equates to the signal contrast divided by any fluctuations in the large background. For shot-noise-limited experiments, these fluctuations amount to $N^{-1/2}$, where N is the number of photoelectrons detected by the camera. Most available cameras exhibit full well capacities on the order of tens to several hundred thousand photoelectrons resulting in shot-noise-induced fluctuations on the order of 1–0.1%. If better sensitivity is required, the magnification of the imaging system can simply be increased to detect more photoelectrons per diffraction-limited spot.⁴⁶ The general rule for optimal localisation in particle tracking recommends an effective pixel size of approximately half the diffraction limit, *i.e.* about 100 nm in the visible.⁴⁷ As an example, a 1000 $\times$ magnification with 10 micron pixels and a pixel illumination yielding 2×10^5 electrons, as is common for CMOS cameras, results in 2×10^7 detected electrons per effective $100 \times 100 \text{ nm}^2$ per pixel and a background noise of only 0.02%. Given that 5 nm AuNPs generate a signal on the order of 0.3%, this would be sufficient for <3 nm localisation of their position and a SNR of 15. Importantly, the requirements for bit depth of the camera are low because any measurements beyond 10-bits simply serve to resolve shot noise. The precision of the measurement is eventually recovered by summing all the counts from the pixels, effectively increasing the dynamic range.

Only one source remains that prevents true shot-noise-limited recording of images: residual scattering from the sample surface caused, for example, by microscope cover glass roughness as is visible in Fig. 3. It is possible to reduce this effect using a multi-

colour approach⁴³ in combination with the specific plasmon resonance of AuNPs, but in most tracking applications it can be removed much more easily: since we are interested in the motion of the particle, the background necessarily is constant and can therefore be very effectively subtracted. Whenever the sample itself is moved, a new background needs to be recorded. However, simple median analysis of any such movie provides exactly this background with exquisite sensitivity making the subtraction very effective^{44,48,49} as shown in Fig. 5.

Overall, the experiment can be thought of as a combination of classic confocal excitation and wide-field detection. One advantage of this approach is that it is trivial to add a simultaneous fluorescence channel by replacing one of the reflective mirrors with a dichroic mirror as shown in Fig. 4b. Taken together, the resulting iSCAT SNR can be written as:

$$\text{SNR} = \frac{\text{iSCAT signal}}{\text{background fluctuations}} = \text{contrast} \sqrt{N} \quad (7)$$

This equation has two very important implications:

(1) The SNR for a given exposure time can be improved at will since $N \propto |E_i|^2 r^2$.

(2) The SNR can be kept constant as the integration time is reduced, again by keeping N constant through increasing $|E_i|^2$.

These are clearly very strong statements, although not at all surprising for a shot-noise-limited experiment. In the section below, we will show that both of these predictions can be easily achieved under realistic experimental conditions leading to extremes in sensitivity and speed that have until recently been deemed far beyond reach in fluorescence-free optical microscopy.

3. Applications of interferometric scattering microscopy

3.1 Lateral single-particle tracking

Single-particle tracking studies have become one of the major applications of optical microscopy in the twenty first century. Interferometric scattering microscopy is not the exception and offers fundamental advantages with respect to single-emitter-based approaches. One representative example is the study of Simian Virus 40 virus-like particles labelled with fluorescent quantum dots on a supported bilayer membrane acquired with the experimental setup from Fig. 4b shown in Fig. 6a. The corresponding simultaneously acquired fluorescence image is shown in Fig. 6b. The signal contrast produced by these 36 nm diameter protein shells is comparable to that of 15 nm AuNPs.

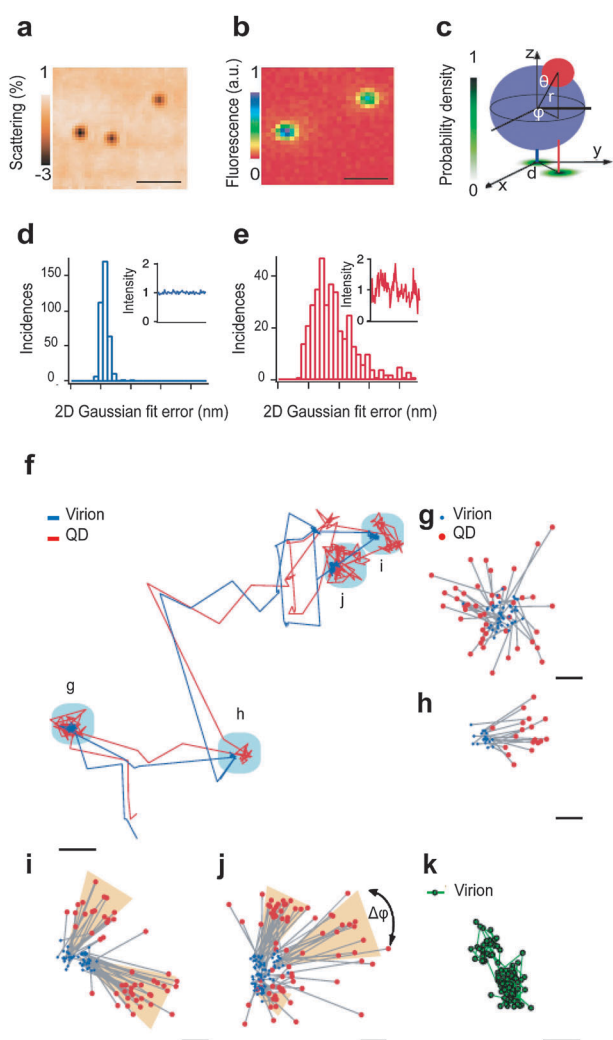


Fig. 6 (a,b) Interferometric scattering and fluorescence image of singly QD-labelled SV40 particles bound to receptor GM1 diffusing on a supported lipid bilayer. Scale bars, 1 μm . (c) Determination of the distance, d , the polar angle, θ , and the azimuthal angle, ϕ , of a virus like particle (blue) labelled with a QD (red) from the scattering and fluorescence centres of mass obtained by localisation (green). (d,e) Fit statistics of a single virus detected by iSCAT and the respective QD attached to the virus detected by fluorescence, respectively. Insets show the normalized signal amplitudes detected by iSCAT and fluorescence, respectively. (f) Representative trajectory of a SV40 particle and its QD label from a sequence of 250 frames acquired by iSCAT with an exposure time of 20 ms. Scale bar, 50 nm. (g–j) Highlighted sections from (f) enlarged and with gray lines connecting the position of the SV40 particle and its QD label for each frame. Orange triangles highlight the confinement of ϕ . Scale bars, 10 nm. (k) Representative rocking motion exhibited by an unlabelled SV40 particle. Reproduced from ref. 45.

The increase in particle size compensates for the reduced refractive index difference with respect to the surrounding water resulting in a comparable scattering signal (see eqn (1)). Surface roughness caused by the cover glass was removed by subtraction of an image recorded after the viruses had diffused out of the field of view.

The scattering contrasts of virions with and without the quantum dot are identical within the error of the measurement since the iSCAT contrast attributed to the QD is $<0.3\%$ and

thus buried in the shot noise of the image. Fluorescence and iSCAT images are registered to within a few nm precision using standard fluorescent beads.⁴⁵

Given that the sizes of the virus, the linker and the quantum dot are all known with high precision, the distance between the centres of mass retrieved from the iSCAT and fluorescence images can be used to extract the orientation of the virus. Nevertheless it remains ambiguous whether the label is above or below the equator of the virus (Fig. 6c). Fitting of the recorded PSFs to 2D Gaussians generates trajectories such as those shown in Fig. 6f. The fit statistics in Fig. 6d and e clearly show the difference between iSCAT and single emitter fluorescence: while the iSCAT signal is stable and the resulting localisation histogram narrow and centred about 2 nm, the fluorescence intensity fluctuates wildly due to photoblinking resulting in an asymmetric and broad histogram of localisation accuracies.

The ability to not only observe the position but also the orientation of the virus as it diffuses on the membrane reveals remarkable nano-scale dynamics. As shown in Fig. 6g–k, the motion of the virus is interrupted by brief, effectively static periods, reminiscent of its behaviour when binding to live cells.⁵⁰ Interestingly, these static periods reveal a range of dynamics from loose (Fig. 6g) to tight (Fig. 6h) and even stepping between close-by nanoscopic spots (Fig. 6i and j), which is also observed in unlabelled virions (Fig. 6k). Together with measurements on giant unilamellar vesicles,⁵⁰ these results suggest rapid exchange between receptors and binding sites on the surface of the virus and hint at the importance of virus and membrane structure for the initial steps of endocytosis.

Apart from the biological significance, these results demonstrate many of the fundamental advantages of iSCAT in terms of a lack of sensitivity to photophysical and photochemical effects. At the same time, they provide a glimpse into a future of making rapid measurements of nanoscopic orientation dynamics beyond what is possible in standard centre of mass single-particle tracking.

3.2 Label-free detection

The previous experiments focused on tracking the absolute motion of a virus particle using iSCAT and fluorescence imaging. Careful examination of eqn (1) suggests that iSCAT can in theory detect nano-objects without the use of any labels, since the scattering amplitude is a function of the difference in refractive index between a nano-object and its surrounding environment. As proof of concept, the dark spots in Fig. 6a are entirely due to the scattering of the virus itself. This can be easily confirmed by comparing the iSCAT signal from QD-labelled and unlabelled virions in the image. In this particular case, the SNR was 30 for an exposure time of only 10 ms. In addition, SV40 is one of the smaller known viruses, and these experiments used virus-like-particles, which lack the packaged viral genome and thus have a smaller refractive index difference and polarisability. This level of sensitivity is only matched by extremely high-Q micro-cavities, which require much more complex experimental setups and sample arrangements.^{51–53}

Beyond its ability to detect nanoscopic objects, iSCAT is also superbly sensitive to refractive index variations. Different

groups have exploited such refractive index changes to study the mechanism of cell adhesion^{37,42,54} or the interaction between viruses and receptors on supported lipid bilayers.⁵⁴ Another more recent example is given by the apparent roughness of a cleaned microscope glass cover slip in iSCAT as shown in Fig. 3. The signal fluctuates by almost 0.5% peak-to-peak even though an AFM image of a similar cover slip reveals roughness on the order of only a few nm RMS over the same area. Thus, while iSCAT does not have the spatial resolution of an AFM, it certainly can be used to detect tiny changes in refractive index on a few nm length scales.

3.3 Using interference for 3D tracking

A natural extension of lateral diffusion is observation of motion in three dimensions. For fluorescence-based techniques, several approaches have been developed such as out-of-focus imaging,⁵⁵ multi-plane detection,^{56–58} astigmatic imaging,^{59,60} PSF shaping with adaptive optics,^{61,62} and interference based methodologies finding the most use.^{39,63} In general, localisation along the optical axis is roughly three times worse than lateral localisation, which is in line with the relative dimensions of a standard diffraction limited spot.

Close inspection of the expression for the detected iSCAT signal in eqn (6) immediately suggests a route to measuring the z -position of a tracked particle: the phase, ϕ . The total phase is a combination of several factors including the Gouy phase, the scattering phase, as well as any phase difference accumulated due to a different path-length of the scattered compared to the reflected light as depicted in Fig. 7a. For perfectly focused imaging, $\phi_{\text{Gouy}} = \pi/2$, while the scattering phase can be calculated based on the refractive index of the material at the wavelength of illumination (0.26π for AuNPs).⁶⁴ For confocal imaging, the variability of the Gouy phase can be used to increase the iSCAT contrast for objects with a low scattering phase, *i.e.* non-resonant scatterers such as viruses or QDs.⁶⁵ In fact, such objects show a clear transition from positive to negative contrast, a behaviour that is regularly observed in microscopes with bright-field illumination.

For a fixed objective–sample distance, however, ϕ_{Gouy} is constant, but the overall phase as a function of particle to interface distance, z , can be written as:

$$\phi(z) = \phi_{\text{Gouy}} + \phi_{\text{scat}} + \frac{4\pi zn}{\lambda}, \quad (8)$$

where λ is the illumination wavelength, n the refractive index of the surrounding material, and ϕ_{scat} the scattering phase.

Thus, if the particle moves relative to the reflecting interface, the signal contrast will vary as $\sin \phi(z)$ making it straightforward to extract the z -position. Importantly, because the technique is interferometric, a complete signal inversion is observed for relatively small vertical motions on the order of $\lambda/4n \sim 100$ nm. Hence, the technique is ideally suited to visualize minute motion (< 100 nm) with nm precision along the optical axis.

An excellent example is provided by a recent study of electrostatic trapping in nanofluidic traps and slits.⁶³ To characterise the potential of the traps, 100 nm gold beads were captured in simple devices such as that shown in Fig. 7b and subsequently imaged using iSCAT (Fig. 7c). Particles that

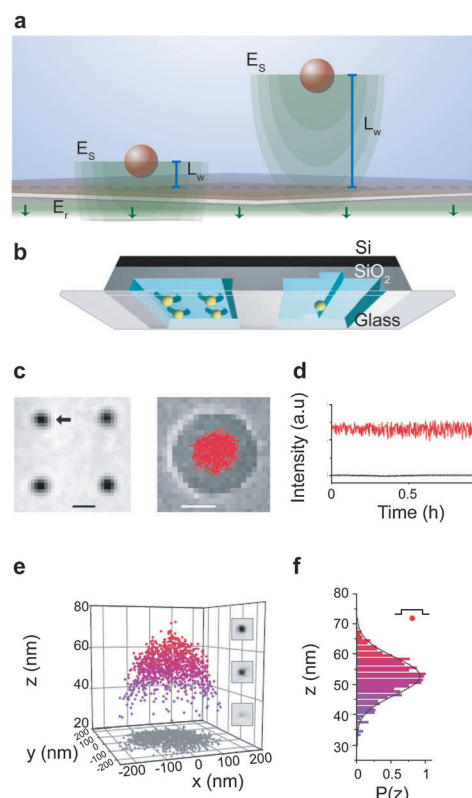


Fig. 7 (a) Conceptual illustration showing the use of interference for three-dimensional tracking of two identical nano-objects in water located at different heights above the surface of a glass coverslip. The difference in optical path lengths (L_w) for each particle leads to changes in the overall phase, which are detected as amplitude changes in the signal generated by the interference between the reflected light field (E_r) and the scattered light field (E_s). (b) Design of fluidic slit devices responsible for electrostatic trapping of nano-particles. (c) Left, representative iSCAT images of confined single 100 nm gold particles. Scale bar, 1 μm . Right, corresponding trajectory of highlighted particle overlaid onto an SEM image of the underlying surface topography. Scale bar, 200 nm. (d) iSCAT signal fluctuations from a confined particle (red) and background (black). (e) Three-dimensional scatter plot of a confined particle with insets depicting representative iSCAT images at different axial displacements. (f) Corresponding axial displacement histogram. Reproduced from ref. 63.

were fixed to the glass surface exhibited a low, but constant iSCAT signal, while the signal from particles that were trapped fluctuated significantly in time (Fig. 7d) when imaged at 1 kHz. Using eqn (8), it was possible to reconstruct the full three-dimensional motion of the particle with nm-precision (Fig. 7e and f) and thereby characterise the potential and trapping mechanism under different conditions. In general, this work demonstrates how 3D-iSCAT in combination with tiny scattering labels can be used to characterise nanoscopic potentials without any limitations to acquisition speed and duration, which has important consequences on the quality of the obtained statistics and resulting potentials. Large particles were used in this case to maximize the electrostatic charge and optimise the trapping potential.

3.3 Ultimate speed: towards the μs barrier

We have already demonstrated a rapid improvement in imaging speed all the way from 0.1 Hz for sample scanning

up to 1 kHz using iSCAT. Nevertheless, even at these frame rates the imaging speed is still limited by the blurring caused by particle motion; for instance, a 10 nm protein freely diffusing in water ($D = 43 \mu\text{m}^2 \text{s}^{-1}$) would have moved on average 500 nm during the integration time of 1 ms; inevitably compromising the precision of localisation.⁶⁶ Therefore a further improvement down to the 10 μs threshold with nm precision is required. However, at such short exposure times, even beam scanning with AODs becomes too slow to illuminate a large area. In addition, full frame acquisition would generate data throughputs in the range of several hundreds of Mega-bytes per second.

To alleviate these difficulties but still enable imaging at tens of kHz we opted for an alternative approach as depicted in Fig. 8a. Here, the incident light is no longer scanned across the sample, but simply illuminates a small area of a few microns by under-filling the microscope objective. The returning light is then imaged onto a fast camera. Despite the non-uniform illumination and subsequent difficulty to identify scatterers in raw images, a simple background correction performed in real time enables quick and easy detection of scatterers. With such a setup, imaging speed is only limited by the speed of the camera and the amount of power one is willing to expose the sample to.

A typical trajectory of a 20 nm particle bound to cholera-toxin on a GM1-doped supported lipid bilayer acquired with this approach is given in Fig. 8b. The frame rate was limited by our camera to 10 kHz while the exposure time was set to the minimum of 10 μs . Nevertheless, single-particle tracking with 2.5 nm lateral precision could still be performed. To demonstrate the dramatic difference in obtained trajectories, we also computed what the same trajectory would look like if recorded in fluorescence at a more standard imaging speed of 100 Hz with the same precision and a similar sized label—a quantum dot (red). Interestingly, while the trajectory looks quite different, the obtained localisation error and full-width at half-maximum (FWHM) of the fitted PSF are indistinguishable (Fig. 8c and d). Under these circumstances, there is no indication that we may have missed brief, albeit important dynamic events. In this particular case, the particle exhibits Brownian motion and thus by itself not too exciting, but the scenario may be rather

different in a wide range of other systems where important dynamic events of a single particle may go unseen, such as in molecular motors.

3.4 Ultimate sensitivity: detecting single molecules without fluorescence

In the same way that it is possible to ever decrease the exposure time while keeping the SNR constant by increasing the incident power, it should also be possible – for a shot noise limited experiment – to increase the exposure time at constant incident power to ever increase the SNR. The ultimate test of such an experiment could be considered to be the detection and imaging of light scattering from a single molecule. One of the major obstacles to overcome is dealing with the signal from the residual background. This challenge is present in the same way as in pure scattering detection but for signals of much smaller magnitude. As an example, the cover glass roughness shown in Fig. 3 produces an iSCAT signal that is two orders of magnitude larger than that expected from a single molecule. Thus, even if the sample moved by only 1 nm during the integration time necessary for detecting a single molecule, the error in background subtraction would overwhelm any signal from the molecule.

One approach to remove such background signals is to create an index matched environment using polymers and microscope immersion oil.^{46,67} In such a scenario, scattering due to the roughness is minimised according to eqn (1), making the detection of weak but resonant scatterers feasible. At the same time, the reduced reflection at the now almost non-existent interface causes the many other reflections present to become significant again, just as in pure scattering detection experiments. One approach to circumvent this problem is to detect the signal in transmission, much like in a standard UV/VIS spectrometer as shown in Fig. 9a. In this way, spurious reflections are negligible again, while scattering due to sample roughness is minimised. In terms of the detected signal, very little changes, except that the reflected field is now replaced by the transmitted field:

$$I_{\text{det}} = |E_t + E_s|^2 = |E_t|^2 \{t^2 + |s|^2 - 2t|s|\sin\phi\} \quad (9)$$

where t takes into account any losses from the sample to the detector. The ratio E_s/E_t is much smaller than E_s/E_r resulting

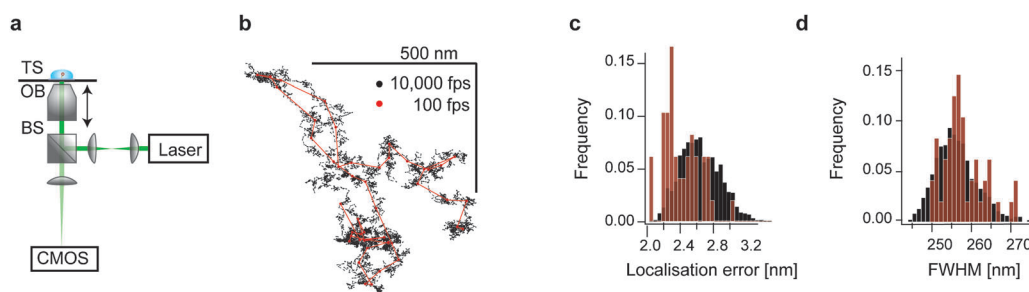


Fig. 8 (a) Schematic diagram of the experimental setup used for fast imaging. A collimated beam at $\lambda = 532 \text{ nm}$ from a solid-state diode laser is focused through an under-filled objective onto a small region of the sample ($2 \times 2 \mu\text{m}^2$). Light is then collected and focused onto a fast CMOS camera for imaging. (b) Reconstructed trajectory of GM1-bound cholera toxin labelled with a single 20 nm Au particle imaged with an exposure time of 10 μs and a constant frame rate of 10000 fps (black). Computed trajectory corresponding to 100 fps, generated by averaging 100 images per time step (red). (c) Corresponding histogram of the retrieved localisation error. (d) Histogram of the obtained FWHM for 10000 images at 10000 fps (black) and corresponding images at 100 fps (red). The combination of the reduced frame rate (100 fps) and localisation precision (2.5 nm) was chosen specifically: it represents the optimum achievable with single QDs.

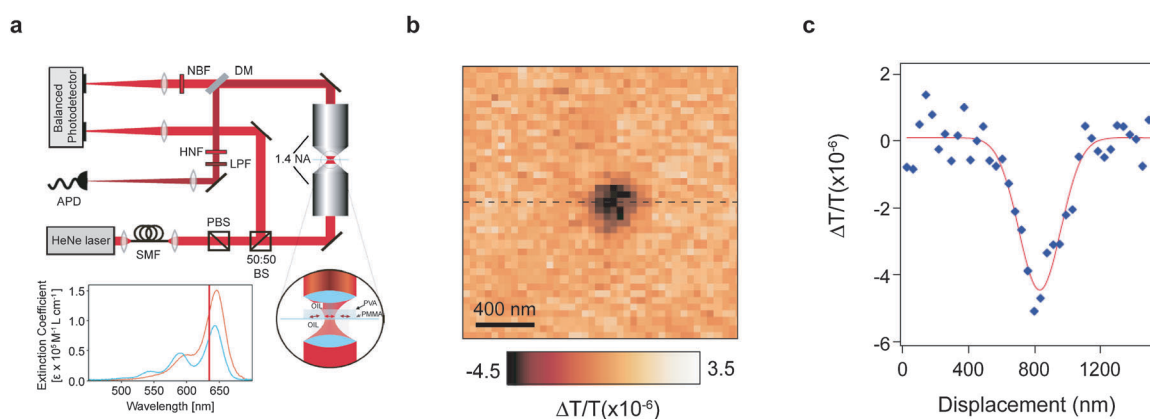


Fig. 9 (a) Top: experimental setup for single molecule absorption. He-Ne: helium-neon; SMF: single-mode polarization-maintaining fiber; PBS: polarizing beam; BS: beam splitter; APD: avalanche photodiode; HNF: holographic notch filter; LPF: long pass filter; DM: dichroic mirror; NBF: narrow band filter; NA: numerical aperture. Inset: sample construct. Bottom left: ensemble absorption spectra for terrylene diimide, TDI, (blue) in toluene and for Atto647N (red) in water. The laser excitation wavelength at 633 nm is indicated for clarity. Bottom right: sample construct. The single molecules studied were either TDI or Atto647N. TDI was embedded in an 80 nm thin layer of polymethylmethacrylate (PMMA) and covered by a 1 μm layer of polyvinyl alcohol (PVA) for protection from microscope immersion medium. Atto647N was spin-cast directly onto a microscope cover glass and then protected by a 1 μm PVA layer. (b) Differential transmission map of a single TDI molecule after subtraction with the corresponding transmission map of the same molecule after photobleaching (average of 10 images each with an integration time of 2 ms per pixel). (c) Cross section marked in (b) with a Gaussian fit indicated by the solid red line. Reproduced from ref. 46.

in a smaller signal contrast. However, as long as the experiment is shot-noise-limited, the SNR is unchanged. Thus, a 20 nm AuNP, which has a 4% contrast in regular iSCAT would now only show a 0.3% contrast. At the same time, the amount of detected light increases 200-fold resulting in an identical SNR. This is a very important point: no matter how the experiment is performed (transmission, reflection, dark-field), if it is shot-noise-limited the SNR is always the same, within a factor of 2. Recall that the SNR for a shot-noise-limited process is given by the square root of the number of photons detected. In the case of pure scattering detection, the number of detected photons is proportional to $|E_i|^2 |s|^2$ which makes the SNR proportional to $|E_i| |s|$. In the interferometric case, the detected signal after background subtraction is proportional to $2\text{Re}[E_r^* E_s] = 2|E_i| |r| |s| \sin(\phi)$ and the noise (ideally in the shot-noise-limited regime) is proportional to the amount of photons detected $|E_i|^2 r^2$ or $|E_i|^2 s^2$ depending on the detection of iSCAT.²⁸ Therefore the SNR for either iSCAT detection mode is proportional to $2|E_i| |s| \sin(\phi)$.

The challenge is usually to ensure that the experiment is shot-noise-limited, and this becomes increasingly difficult as the detected signal becomes smaller. This is the intrinsic reason why detecting in reflection mode is generally preferable: reducing E_r increases the contrast and thus demands background fluctuations on the order of 0.1% which is relatively easy to achieve with both CCD cameras and photodiodes. When measuring in transmission mode, however, background fluctuations due to instrumentation and to residual scattering from sample imperfections and changes in refractive index have to be suppressed to a much larger degree, for instance below the parts-per-million level to detect single molecules.

With the described experimental setup, the only significant remaining noise source arises from laser intensity fluctuations. Because these experiments are performed with mW input powers and CW laser beams, balanced detection can be used

to suppress laser noise. Such detectors generally exhibit common noise rejection on the order of 50–60 dB making the required sensitivity possible with lasers that exhibit noise levels of 0.1% in the required detection bandwidth, which are commonly available.

Overall, this approach not only allows clear extinction detection, but also the imaging of single molecules as shown in Fig. 9b. Using multiple laser colours, even spectroscopy is feasible in the future. We remark that fluorescence-free detection of a single molecule has recently also been achieved using ground-state depletion and photothermal detection.^{68,69} Because the iSCAT approach is the most direct measurement, however, it has produced the highest SNR. Finally, since it does not necessarily rely on any resonant interaction, the extreme sensitivity can also be used for label-free detection down to the single molecule level. In fact, an average sized (5 nm) individual protein is expected to generate roughly the same iSCAT signal as the dye molecules used above, paving the way towards true label-free single-molecule detection.

4. Outlook

A natural question that arises from these experiments is: how far can we go in terms of sensitivity and imaging speed? We have repeatedly made the argument, and shown experimentally, that the final limiting factor is always shot noise. In this respect, the sensitivity and speed limits are ultimately set by the amount of light one can expose the sample to. For single dye molecules, photobleaching is a fundamental problem, which limits the number of photons that can actually interact with the molecule. This in turn restricts the photon flux and eventually the detection bandwidth with iSCAT to about 0.1 Hz. However, molecules that do not exhibit a high fluorescence quantum yield together with a ns excited state lifetime, but rather decay rapidly out of the excited electronic state, such as quenchers,

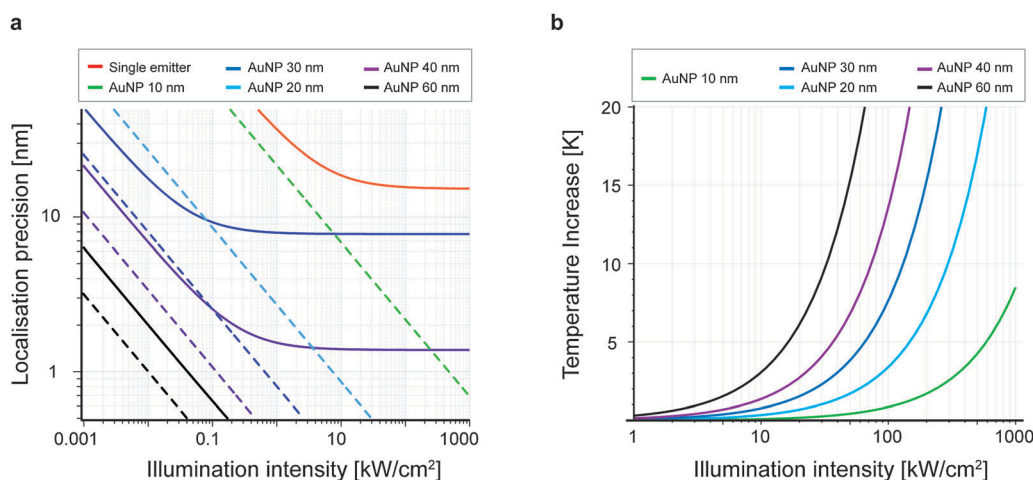


Fig. 10 (a) Localisation precision dependence on incident illumination from different techniques based on the current literature for 100 μ s exposure time: dark-field microscopy (solid lines) and interferometric scattering microscopy (dashed lines). (b) Theoretically calculated maximum temperature increase on the surface of a single spherical nano-particle induced by absorption of incident light at 532 nm.⁷²

can handle much higher input intensities without being subject to damaging photochemistry.⁶⁸ For such molecules, or even non-resonant ones, much higher input powers are possible that can easily reduce the detection bandwidth by several orders of magnitude. It will thus be absolutely feasible to perform absorption spectroscopy of single molecules in the near future.

Similar arguments relate to imaging speed and label size. We have already demonstrated tracking of a 20 nm AuNP with 10 μ s exposure time and nm localisation precision. One limitation that will eventually appear is heating of the nanoscopic scatterer as invariably some light is absorbed, especially if the interaction is resonant.^{70–72} To assess the importance of this effect, we have computed the localisation precision as a function of illumination intensity for an exposure time of 100 μ s (Fig. 10a) to enable a meaningful comparison with other available techniques. We have also included corresponding plots for dark-field detection and single-emitter fluorescence based on state-of-the-art experimental setups.^{27,22,73} Dark-field localisation curves level out at the maximum achievable SNR for a given particle size. While 40 nm AuNPs can still be localised to a few nm. We assumed the limiting factor to be the contribution from background scatterers and spurious reflections as presented in state-of-the-art experimental setups.^{27,22,73} These contributions lead to an increase in noise, which then reduces the achievable precision for 30 nm AuNPs and makes 20 nm AuNPs effectively useless for localization measurements. The photon yield and thus localisation precision for dye molecules is limited by saturation and thus levels out at an even earlier stage at such a short exposure time. For iSCAT, however, this asymptotic effect is not observed.

The important question that remains is the associated temperature rise for a given input power. To go to the extreme, localisation of a 10 nm particle with 1 nm precision would incur a temperature rise of 5 K on the surface of the particle. Such an increase appears to be acceptable in most circumstances, especially if it is taken into account that it drops off inversely with the distance from the surface. Overall, Fig. 10a and b can be used as a general guide. For 10 μ s exposure time with 1 nm precision and a 20 nm AuNP one would require an

incident intensity of $\sim 70 \text{ kW cm}^{-2}$ which in turn would result in surface heating on the order of $\sim 3 \text{ K}$.

Can one improve even beyond these limits? In principle, there are two possibilities. Firstly, one could use non-resonant particles such as silica spheres that cause little heating, but in exchange require even higher powers due to the smaller scattering cross sections. On the other hand, silver nano-particles (AgNPs) are an alternative. Due to their sharper and stronger plasmon resonance AgNPs provide roughly the same scattering properties at 400 nm as gold at 532 nm, but with roughly half the particle size, making experiments with 5 nm particles feasible.

We have given a comprehensive review of iSCAT, both from an experimental and theoretical perspective and outlined the strengths and weaknesses of the technique. Due to its sensitivity to any scattering background, experiments in live cells are likely to always favour more background-free approaches such as fluorescence or photothermal imaging. At the same time, however, this sensitivity can be exploited for label-free imaging and detection. For *in vitro* experiments in low scattering environments, the speed and sensitivity of iSCAT are likely to make it the high-speed camera of the nano-world. In the same way that we are now used to seeing movies of exploding airbags, iSCAT will be providing the same type of movies, but of nanoscopic motion. The tremendous increase in imaging speed of 3–4 orders of magnitude compared to single emitter-based techniques and its much improved match with the speed of nanoscopic motion under physiological conditions create a bright future for iSCAT in both the life sciences and condensed matter physics.

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